

Bias, Fairness, Surveillance, and NLP

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Learning Goals

- Getting an understanding of how the field has thought about bias and fairness
- And how those are limited in their scope
- Get acquainted with the political project of surveillance and how machine learning is contributing



The Eye of Sauron

“AI and technology can be transformative to the [...] law and order space [...] When I was in justice, my ultimate vision for [...] the criminal justice system was to achieve, **by means of AI** [...], what Jeremy Bentham tried to do with his Panopticon. **That is that the eyes of the state can be on you at all times.** I think there’s big space here for being able to harness the power of AI and tech to get **ahead of the criminals**”
- Shabana Mahmood (UK Home Secretary)



Part I

What even is “bias”



Social Impact Categories

- Systems
- People and Society



Base System

- Biases, Stereotypes, Representational Harms
- Cultural Values and Sensitive Content
- Disparate Performance
- Privacy and Data Protection
- Environmental Cost and Carbon Emissions
- Labor Impact
- Financial Costs



Base System

Representational Harm

E.g., representing Muslims as terrorists

Allocational Harm

E.g., when systems are used to make decisions which



An Epistemology of Bias/Fairness

- A mathematical epistemology of bias and fairness
- What does a mathematical epistemology imply?



An Epistemology of Fairness

- A mathematical epistemology of fairness
- What does a mathematical epistemology of fairness imply?
 - Only things that are countable can be understood to impact fairness
- The outcome here is that we can do things like operationalize utilitarianism



The trouble with defining “bias”

- The limitations of quantifiability of “bias”
- “bias” is necessarily context dependent
- And is only meaningful in the context of marginalisation
- ML is for minimising the expectation of error
- And is built on human, political data
- Debiasing as a political act



Defining “bias”

***Def 1:** Bias is the existence of an undesirable position with some imagination of a desirable position.*



Defining “bias”

***Def 1:** Bias is the existence of an undesirable position with some imagination of a desirable position.*

***Def 2:** Bias is the systematic undesirable position produced with regard to existing systems of oppression.*



Part II

Addressing Biases



Conceptual Uncertainty

- Poorly defined definitions



Conceptual Uncertainty

- Poorly defined definitions
- Validity Issues



Measurement Validity

Measurement model

A measurement model is a model that tries to infer unobservable concepts from observable and non-observable attributes



Measurement Validity

- *Construct Reliability*: Is the construct itself valid?
- *Construct Validity*
 - *Face validity*: Does it intuitively make sense at all?
 - *Content validity*:
 - *Contestation*: Is there contestation around the construct?
 - *Substantive validity*: Are **only** relevant properties used?
 - *Structural Validity*: Is there a relationship between properties and the construct?



Measurement Validity

- *Convergent Validity*
 - Is there a relation with existing measurements?
- *Discriminant Validity:*
 - Are some parts of the operationalisation shared w/ other constructs ?
- *Predictive Validity:*
 - Are the results from a measurement model predictive of its constituent constructs?



Measurement Validity

- *Hypothesis Validity*
 - Can we draw interesting and meaningful hypotheses from the outcomes of the measurement model?
- *Consequential Validity*



Conceptual Uncertainty

- Poorly defined definitions
- Validity Issues



Conceptual Uncertainty

- Poorly defined definitions of Bias
- Validity Issues
- Power, Control, and Agency
 - To define, implement, subject



Part III

NLP and

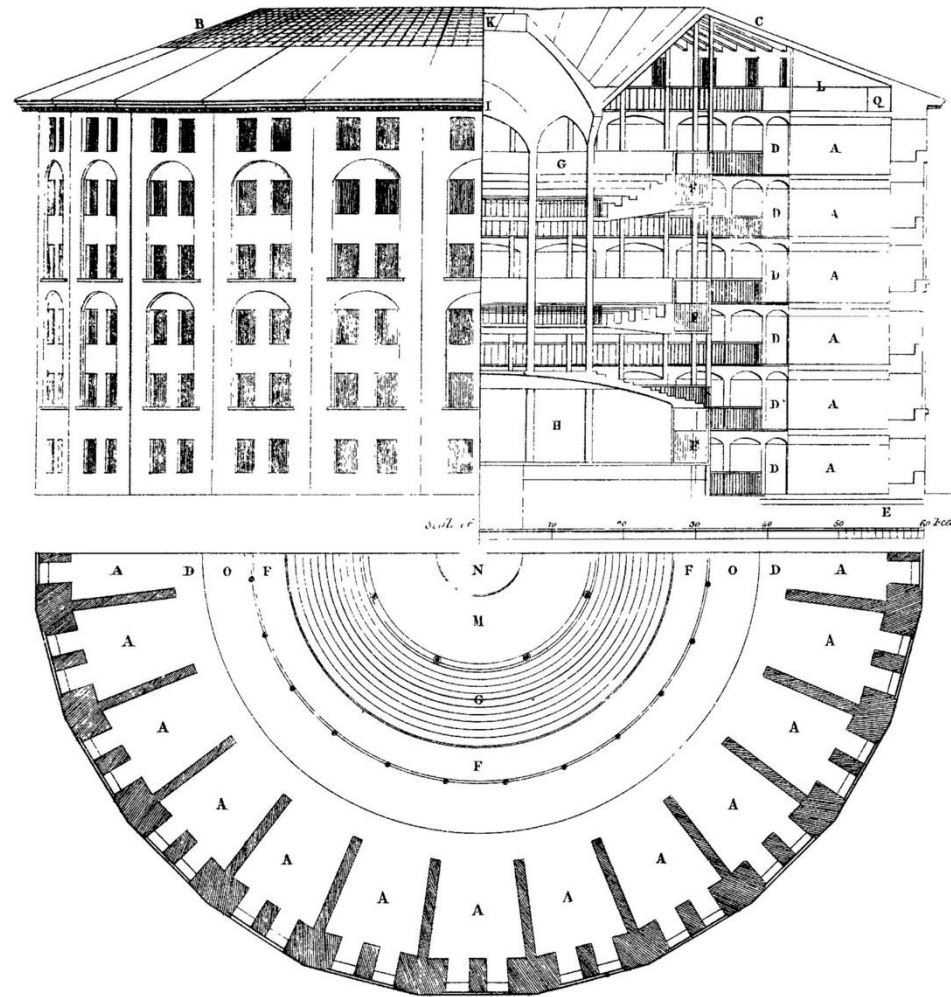
Surveillance



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By Jeremy Bentham - The works of Jeremy Bentham vol. IV, 172-3, Public Domain.

A (super) brief history of the Internet

- 1969-1986: ARPANet & NSFNet
 - Connected states in the US, the UK, and Norway
 - Connected NSF supercomputers of the time across US universities across states and universities in Europe
- 1989/1990: The World Wide Web
 - Connecting all websites with each other
 - Online advertisement
- 2004–now: Web 2.0



The Social Web (2.0)

- Web 2.0: User-generated data
- Rise of Facebook, Twitter, YouTube, etc.
- Mobiles phones and wireless devices
- Google introduces Google Ads



Collecting Data

- Collecting data from people's posts
- Advertisement Networks and tracking cookies



Data on data

- Large data collections
 - User-contributed texts, images, and videos.
- Finding the needle in the haystack
 - Instantiating the surveillance network with data analysis
 - ... And machine learning (eventually)



Sifting through the haystack

- 2010s: ML works for text
 - Specifically English
 - Starts to work for vision
- 2020s: Multilingual ML
 - For some.



Affordances: A brief Interlude

- An affordance is a potential action or outcome that is made available by a thing
 - A small flat stone affords being tossed into a lake to skim it on the surface
 - A large boulder does not afford being thrown anywhere



Surveillance

- Watching from above.
- 3rd party performing the act of watching
- The surveilled does not have an option to opt out



A State of Surveillance?

- Internet-facilitated tracking
- Recorded longitudinal data
- Large-scale data processing capabilities



Surveillance and Bias

- Accuracy of models



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